

O-ISAC: Supplementary Evidence Profiles

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These profiles accompany the systematic review of 206 studies represented by 227 eligible reports. They describe the distribution and interpretation of the extracted evidence. The separate supplementary methods narrative documents search, selection, extraction, synthesis, appraisal and limitations. The source index links these materials to the complete study register and frozen evidence tables in the review archive: <https://osf.io/7f6wb/>.

1 S-Core: Platform, Metric, and Relationship Evidence Profiles

1.1 Platform Assignment

The main-text platform table retains the canonical mutually exclusive family counts for 206 studies. Each study is assigned to its dominant physical platform, while physical media and integration mechanisms can overlap within the same system. Transmitters supplied through fiber enter the photonic terahertz family when photonic generation and conversion create the wireless carrier directed at the target. Fiber remains the organizing platform when it transports signals to downstream RFID, radio, or radar sensing. Hybrid membership requires interaction across segments to shape the result. Family counts describe corpus coverage rather than relative performance or maturity.

1.2 Metric Evidence

The primary synthesis contains 4,779 metric records. A record is an extraction unit preserving a reported quantity, its available definition, conditions, and source location; it is not an independent study or effect estimate. Data rate, BER, and range resolution appear in 85, 60, and 50 studies, respectively, with overlapping study coverage.

Table S1: Domain coverage of the primary metric records

Domain	Records, n (%)	Studies, n (%)
Sensing	1,816 (38.0%)	203 (98.5%)
Communication	1,328 (27.8%)	194 (94.2%)
Joint	870 (18.2%)	158 (76.7%)
Implementation	476 (10.0%)	64 (31.1%)
Smaller normalized domains	289 (6.0%)	33 (16.0%)

Domain rows partition the primary metric records, whereas study counts overlap. Of these records, 118 (2.5%) are conditional comparison candidates and 4,661 (97.5%) retain their native-context analytical role. Candidate status requires the reported definition, task, measurement plane, operating conditions, and validation to be checked before any proposed comparison. It does not establish a verified comparison across independent studies or justify pooling. The remaining records support mechanism, feasibility, and qualitative synthesis in their source contexts. The complete metric register contains 4,861 rows, including 31 contextual records and 51 records documenting source conflicts. Electronic Supplement S-Evidence preserves the complete records and provenance.

1.3 Joint Performance Relationships

The relationship register contains 404 entries. Two document the absence of a tradeoff, leaving 402 substantive relationships from 168 studies. These include 218 quantitative and 184 qualitative records; 371 substantive relationships

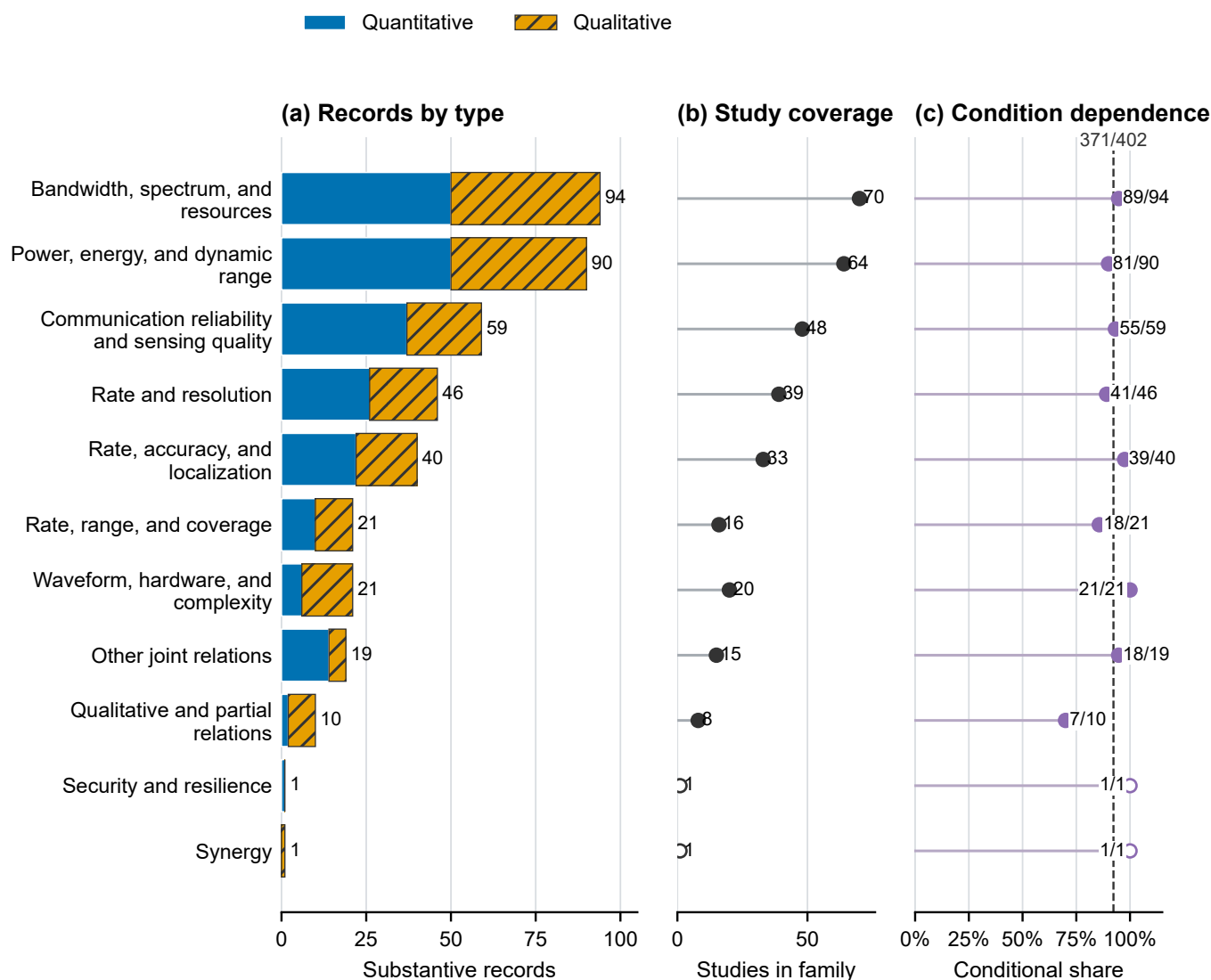


Figure S1: Descriptive profile of the 402 substantive joint-performance relationships. Panel (a) separates quantitative and qualitative records; panel (b) reports overlapping study coverage across mechanism families; panel (c) reports condition dependence. Family frequency describes corpus coverage, not effect size or comparative maturity.

have condition-dependent interpretations. The profile in Fig. S1 preserves the overlapping mechanism-family coverage. A relationship is not a study, a pooled effect, or an independently verified cross-study comparison. The main text uses selected within-study examples to explain how a shared choice changes communication and sensing.

2 Validation Profiles and Application Coverage

This supplement preserves the descriptive profiles that support the main article's discussion of joint validation, application requirements, and research priorities. Counts refer to the 206 included studies unless another unit is specified. Overlapping categories describe coded coverage rather than independent replications, market prevalence, or a ranking of platforms.

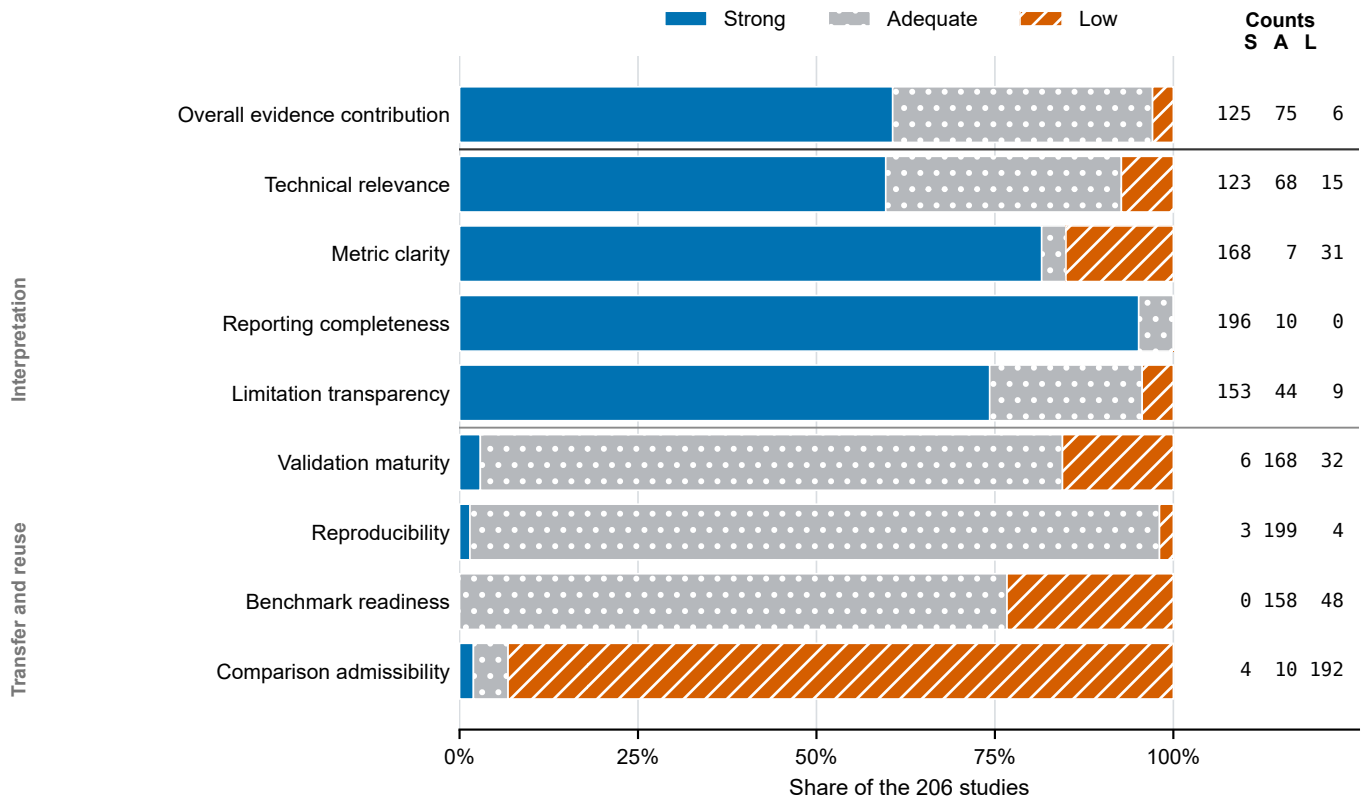


Figure S2: Full review-specific TQAF profile. The eight dimensions characterize support for interpretation, transfer, and reuse under the stated scoring rules. Overall evidence contribution is shown separately; it is not a ninth appraisal dimension. These descriptive scores do not establish a universal study ranking.

2.1 Technical Appraisal and Validation Methods

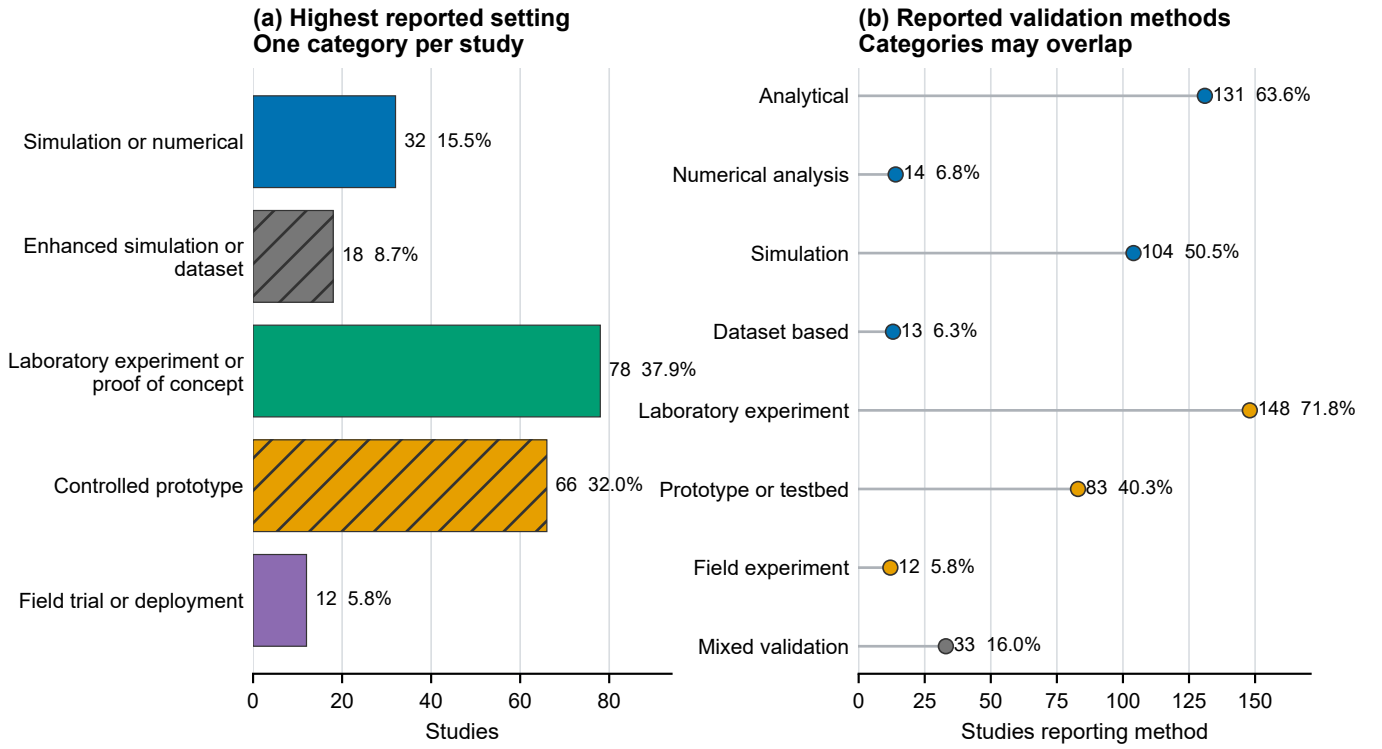
The Technical Quality Assessment Framework (TQAF) scores the eight technical and reporting dimensions in Fig. S2 from 0 to 3, with overall evidence contribution recorded separately. This descriptive appraisal was developed for the review and has not been independently validated; it is neither a conventional risk-of-bias instrument nor GRADE. Scoring rules and study-level records are available in S-Protocol and S-Appraisal, with conduct and interpretation limits in the methods supplement.

The reproducibility assessment combines parameter reporting with artifact access: 4 studies were classified as low, 199 as adequate, and 3 as strong. The benchmark-readiness assessment classified 48 as low, 158 as adequate, and none as strong. Under these review-specific criteria, strong benchmark readiness requires an external or common baseline, an open artifact, admissible outcomes in both functional domains, and strong validation and reproducibility support within the same evidence chain. These criteria are more restrictive than either reporting open data or demonstrating a physical prototype alone.

The highest-setting classes contain 32 simulation or numerical studies, 18 enhanced-simulation or dataset studies, 78 laboratory or proof-of-concept studies, 66 controlled prototypes, and 12 field or deployment studies. Six of the 12 field studies report outcomes in both communication and sensing at field or deployment level. This study-level classification does not establish concurrent operation in one configuration or time interval. S7 preserves separate source locators for the two functions and the available relationship timing.

2.2 Technology and 6G Relevance Inventory

Table S2 preserves the original overlapping technology counts and the exclusive 6G relevance classification. Technology labels can occupy different positions in the same signal chain, so their frequencies do not represent competing architectures or a universal sequence. For example, photonic integration occurs in 20 studies, distributed acoustic sensing in 22, machine learning or artificial intelligence in 20, and digital twins in two. Nineteen studies include specialized



Settings describe how far a study reached. Methods describe the evidence used along that path.

Figure S3: Validation settings and overlapping evaluation methods. Panel (a) assigns each study one highest reported validation setting. Panel (b) permits overlapping methods: analysis, numerical or dataset-based work, simulation, laboratory experiments, prototypes, field studies, and mixed methods may contribute to the same study. Method counts must therefore not be added to infer a number of independent studies or experimental replications.

Table S2: Technology-label counts and 6G relevance classes

Label	Studies, n
<i>Technology labels: overlapping</i>	
Photonic terahertz generation	68
Chirped processing	66
Coherent optics	64
Multicarrier waveform	56
Fiber sensing reuse	22
Photonic integration	20
Machine learning	20
Other specialized technologies	19
Beamforming	13
Optical phased array	8
Multiple input multiple output	7
Digital twin	2
Optical intelligent surface	1
<i>6G relevance: exclusive</i>	
Direct relevance	138
Inferential relevance	64
Weak relevance	1
Not applicable	3

mechanisms grouped under other technology labels. These categories capture device integration, infrastructure reuse, inference, and additional specialized mechanisms.

Direct 6G relevance means that a report explicitly connects its system to a 6G role; inferential relevance identifies an enabling contribution requiring additional network context. The remaining classes describe lighter connections or application settings outside that framing. The application category “6G and access” in Table S3 is a separate,

Table S3: Application domains and operating requirements in the reviewed corpus

<i>Network and mobility</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
6G and access 100 (48.5%)	Access and transport with channel, target, position, or environment sensing	Timing, orchestration, continuity, and evidence for both functions	[1, 2]
Vehicular 41 (19.9%)	Vehicle data links with range, motion, pose, temperature, or channel sensing	Changing geometry, acquisition, tracking, and service continuity	[3, 4]
Optical access 32 (15.5%)	Access and transport links with channel, event, or fiber state sensing	Reach, synchronization, and traffic continuity	[2, 5, 6]
Indoor positioning 13 (6.3%)	Indoor optical links with user or robot position and orientation sensing	Geometry, calibration, coverage, and update continuity	[7, 8]
Data center 4 (1.9%)	Short reach data links with channel estimation or link state sensing	Traffic continuity and calibrated link references	[5]
<i>Infrastructure</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
Environmental monitoring 55 (26.7%)	Distributed links with vibration, temperature, or environmental event sensing	Persistent observation, localization, and traceable reference data	[6, 9, 10]
Smart infrastructure 41 (19.9%)	Infrastructure links with structural, motion, or event monitoring	Continuous observation, calibration, and resilient alert delivery	[1, 6, 11]
Industrial 34 (16.5%)	Industrial data or control with process, strain, temperature, motion, or position sensing	Delay, continuity, calibration, and safe operation	[1, 12]
Security and surveillance 26 (12.6%)	Monitoring links with anomaly, event, or location sensing	Coverage, event discrimination, and alert latency	[6]
<i>Human use, extreme settings, and remaining domains</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
Healthcare 6 (2.9%)	Wearable or ambient links with vital sign, activity, or gesture sensing	Motion robustness, physiological reference, illumination, and user safety	[12, 13]
Aerospace 20 (9.7%)	Satellite links with atmospheric or wavefront sensing	Pointing, turbulence, delay, and link adaptation	[14]
Underwater 2 (1.0%)	Underwater link concepts and subsea data links with temperature or event sensing	Water attenuation or cable calibration, packaging, and access	[10, 11]
Other 17 (8.3%)	Emerging links with range, imaging, object, or channel sensing	Metrics and reporting specific to the task	[15]

Note. Application domains overlap, so a study may contribute to more than one row. Counts and percentages describe coverage within the 206 included studies. For each domain, the table includes a selected study set and lists the corresponding references in the final column. The 6G and access row identifies the application category, and the preceding subsection provides the separate 6G relevance classification.

overlapping application label. Neither classification demonstrates a network-level operating point merely through the presence of a 6G label.

2.3 Complete Application Inventory

Table S3 preserves all thirteen application domains and their study counts, conditions, and selected references. The main article groups these requirements by the experiment needed to test joint operation; that presentation does not alter the underlying taxonomy.

2.4 Metric Records Requiring Conservative Interpretation

The primary metric collection contains 4,779 records, each representing a reported metric observation rather than an independent study or joint effect. Within this collection, 118 records are conditional comparison candidates. Their candidate status does not establish 118 verified cross-study relations, independent replications, or pooled effects. Compatibility still depends on the task, quantity, measurement plane, operating conditions, and source-level evidence, as explained in the main article.

For 92 metric rows, the available source fields supported interpretation in the original setting with a conservative

comparison status. These rows are retained with their source context; missing comparison information is not imputed, and their count does not define an effect estimate. S-Evidence and the metric records preserve the detailed field values and source locators. Artifact-access classifications combine source reporting with targeted link checks and describe the documented access route, rather than the completeness of an independently executed reproduction.

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